VITyarthi | open source software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5

Max Marks: 100

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1. Introduction

The digital world we live in today is built on shared knowledge and collaboration. This report explores both the practical setup and deeper understanding of MySQL on a Linux system.

MySQL is an open-source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL) to store, organize, and manage large amounts of data. It works by arranging data into related tables, making it easy to retrieve and manipulate information efficiently.

Since its creation in 1995 by a small group of Swedish developers, MySQL has grown into one of the core technologies powering the internet. Over time, it has evolved alongside the broader shift from the early “Free Software” movement to its current development and management under Oracle Corporation.

This report not only documents the technical deployment of MySQL on Linux but also reflects on its journey, impact, and role in modern computing.

2. Choosing Software

Approved software options

Software	Category	License	Why it's interesting
Linux Kernel	Operating System	GPL v2	*The foundation everything else runs on*
Apache HTTP Server	Web Server	Apache 2.0	*Powers ~30% of all websites globally*
MySQL	Database	GPL v2 / Commercial	*A license with a dual-model story to tell*
Firefox	Web Browser	MPL 2.0	*A nonprofit fighting for an open web*
VLC Media Player	Multimedia	LGPL / GPL	*Plays anything — built by students in Paris*
LibreOffice	Office Suite	MPL 2.0	*Born from a community fork — a real lesson*
Python	Programming Language	PSF License	*A language shaped entirely by community*
Git	Version Control	GPL v2	*The tool Linus built when proprietary failed him*

Your choice	Write your chosen software here before submission: __MySQL_____

3. The Report

Part A — Origin and Philosophy

A1. The problem this software was created to solve

Structured Query Language (SQL) was originally created to solve a fundamental problem: how to easily access and navigate data stored in computers. Before SQL, retrieving information was a complicated process. Programmers had to guide the computer step-by-step—almost like giving GPS directions—just to find a single piece of data. SQL changed this by introducing a simple "query" approach, where users could write instructions in an English-like format to access information more easily.

The roots of SQL trace back to the early 1970s at IBM's San Jose Research Laboratory. At that time, data was stored in hierarchical or network databases, which followed rigid structures similar to trees. To retrieve something as simple as a customer's address, a programmer had to manually navigate through multiple levels—starting from a "State" node, moving to "City," and then searching through customer records. Even small changes in the data structure, such as adding a new layer, could break existing programs. Additionally, there was no clear separation between how data was logically understood and how it was physically stored. This meant programmers needed deep knowledge of hardware, while non-technical users had no direct way to access data and had to rely on specialists, often waiting days for results.

In 1970, Dr. Edgar F. "Ted" Codd introduced a revolutionary idea through his paper *"A Relational Model of Data for Large Shared Data Banks."* He proposed organizing data into tables with rows and columns, independent of physical storage. Building on this concept, IBM researchers Donald D. Chamberlin and Raymond F. Boyce developed a language called SEQUEL (Structured English Query Language), which was later shortened to SQL. Although IBM was slow to commercialize the idea, Larry Ellison recognized its potential and launched Oracle V2 in 1979, the first commercial relational database system.

Over time, Oracle's implementation of SQL became powerful but also complex and resource-heavy. By the mid-1990s, developers Michael "Monty" Widenius, David Axmark, and Allan Larsson sought a faster and more flexible alternative, especially for the growing needs of the World Wide Web. This led to the creation of MySQL, named after Monty Widenius's daughter, My. Initially, they planned to use the mSQL database system with fast low-level ISAM routines, but when it proved insufficient, they developed a new system with a similar API for easy compatibility. This made it simple for developers to transition from mSQL to MySQL.

MySQL was developed by MySQL AB, a Swedish company founded in 1995, and quickly became a key component of the popular LAMP stack. In 2008, the company was acquired by Sun Microsystems for $1 billion.

A major reason for MySQL's success was its open-source model. Monty Widenius and his team believed that sharing the source code would allow developers worldwide to contribute, identify bugs, and improve performance. This global collaboration made MySQL more stable and efficient than what a small private team could achieve alone.

At the same time, MySQL followed a smart dual-licensing strategy. Open-source projects could use it for free under the GPL license, while companies wanting to integrate it into proprietary

systems had to purchase a commercial license. This approach allowed MySQL to grow rapidly while still generating revenue.

Beyond business strategy, there was also a strong philosophical belief behind keeping MySQL open source. Monty Widenius emphasized transparency and practicality. Unlike some proprietary software that hides its inner workings, MySQL openly demonstrated its efficiency and design. It was a clear statement: if the software performs well, it is because it has been built well, not because anything is hidden.

A2. The license — what it actually says

What are the four freedoms of free software? Does your software's license grant all four? The four freedoms of free software are-

Freedom 0 - The Freedom to Run: The freedom to run the program as you wish, for any purpose. There are no licensing restrictions that prevent you from using the software for specific activities. You don't need to notify the developers or any entity about how you are using it.

Freedom 1-The Freedom to Study and Adapt: The freedom to study how the program works, and change it so it does your computing as you wish. Access to the **source code** is a precondition for this. Without the code, you cannot understand the "logic" of the software or verify what it is doing with your data.

Freedom 2-The Freedom to Redistribute: The freedom to redistribute copies so you can help others. You are allowed to give away or sell copies of the software (binary or source) to anyone, anywhere. This freedom ensures that the software can spread and be accessible to those who need it.

Freedom 3-The Freedom to Improve and Release: The freedom to distribute copies of your modified versions to others. By doing this, you can give the whole community a chance to benefit from your changes. Access to the source code is also a precondition for this.

The MySQL license provides all four of these freedoms. This is reflected in statements such as:

- “The act of running the Program is not restricted” (lines 142–143),
- “You may modify your copy or copies of the Program…” (lines 159–161),
- “You may charge a fee for the physical act of transferring a copy…” (lines 156–157),
- “You must cause any work that you distribute or publish…” (line 167).

These clauses collectively ensure that users can run, study, modify, and distribute the software freely.

Can a company take this software, modify it, and sell it without sharing their changes? Why or why not? A company is allowed to modify and sell the software, but it cannot keep its changes private. According to the license:

- “You must cause any work that you distribute or publish… to be licensed… under the terms of this License” (lines 167–170),
- “You may copy and distribute the Program… in object code or executable form…” (lines 203–205).

This means any modified version must also be released under the same GPL license. Companies cannot hide their changes or keep them proprietary; they must provide access to the source code so users can view and use the modifications.

What is the difference between GPL and MIT licensing? Which would you choose for something you built, and why? The key difference between the GPL (General Public License) and the MIT License lies in their philosophy:

- **GPL License:** It is a "copyleft" license that ensures software remains free and open. Any modifications must also be shared under the same license. Its goal is to protect user freedom and keep software open for everyone.
- **MIT License:** It is a permissive license that gives developers more flexibility. They can modify the software and even keep their changes private. It focuses more on developer freedom than enforcing openness.

If choosing between the two, the decision depends on the goal of the project. GPL is ideal when the aim is to ensure long-term openness and transparency, as seen in MySQL. On the other hand, MIT is better for maximizing adoption and reducing restrictions for developers and companies. Personally, choosing GPL aligns with promoting transparency and keeping software accessible to everyone.

Is there a difference between 'free as in free beer' and 'free as in freedom'? Give an example from your chosen software.

- "Free as in free beer" means the software is available at no cost.
- "Free as in freedom" means users have the rights to use, modify, and distribute the software.

In the MySQL license, both ideas are present. For example:

- "You may charge a fee for the physical act of transferring a copy…" shows that the software is not necessarily free of cost.
- "When we speak of free software, we are referring to freedom, not price…" emphasizes that the real focus is on user rights—such as copying, modifying, and distributing the software.

This clearly highlights that MySQL represents "free as in freedom," not just "free as in price."

A3. The ethics of open source

Should all software be open source? Make an argument for and against.

Open source is more than just a type of license—it reflects a philosophy about how knowledge and tools should be shared openly with everyone.

From a technical point of view, almost any software *can* be made open source, allowing users to access and study its source code freely. However, in reality, many applications remain closed due to legal, business, and security reasons.

One major factor is privacy and intellectual property. Companies like Microsoft and Adobe keep their source code private because it represents their core business value. If they made it public, competitors could easily copy their features and create free alternatives to their paid products.

There are also security concerns. Software used in critical areas—such as power grids, defense systems, or government communication—is kept closed to prevent misuse or access by unauthorized or foreign entities.

So while open source promotes transparency and collaboration, practical considerations like business protection and security often require software to remain closed.

Is it ethical for a company to build a profitable product entirely on top of free community labor? In my view, it depends on how the company uses and contributes back to that community work. If a company is simply taking open-source code, building a product, and charging users without adding meaningful value or giving anything back, it can feel unethical—because they are profiting from others' unpaid efforts.

However, if the company provides additional value—such as improved features, stability, support, security, or enterprise-level services—then it becomes more reasonable. In such cases, they are not selling the code itself, but the reliability and services around it. A good example is MySQL, which is open source and free, but also offers paid features and enterprise support developed by professional teams. Similarly, companies like Red Hat, Google, and Meta use Linux, improve it, and then monetize services like support and infrastructure rather than the code itself.

So overall, it is ethical when companies contribute, add value, or support the ecosystem—not when they simply exploit it.

What does 'standing on the shoulders of giants' mean in the context of software? Does open source enable or undermine original innovation? In software, "standing on the shoulders of giants" means building new technologies using the strong foundation of existing tools, code, and knowledge. Instead of starting from scratch, developers can use established systems like the Linux kernel, MySQL, or Python to create more advanced solutions quickly.

Open source strongly enables innovation rather than limiting it. Because so much high-quality code is already available, developers can focus on solving new problems instead of reinventing basic systems. This lowers barriers to entry—allowing startups, students, and independent developers to build powerful applications without needing huge resources.

For example, a small team can now create a full product in weeks using open-source tools that once required massive investment. This access empowers more people to innovate, making technology development faster, more inclusive, and more creative.

Part B – Linux Footprint

Setting up WSL2 - on terminal: wsl –install. After this we decide our user name and password. To update the package : sudo apt update.

Installation of MySQL- on ubantu: sudo apt install mysql-server -y

To start the database service : sudo service mysql start

key directories

The Configuration Directory: /etc/mysql/

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ls -ld /etc/mysql
drwxr-xr-x 4 root root 4096 Mar 20 09:12 /etc/mysql
hpmuskansukhija@DESKTOP-3TC5UPL:~$
```

The Data Directory: /var/lib/mysql/

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ls -ld /var/lib/mysql
drwx------ 7 mysql mysql 4096 Mar 30 13:15 /var/lib/mysql
```

The Log Directory: /var/log/mysql/

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ls -ld /var/log/mysql
drwxr-x--- 2 mysql adm  4096 Mar 30 13:15 /var/log/mysql
```

The Binary/Executable Directory: ls -l /usr/bin/mysql

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ls -l /usr/bin/mysql
-rwxr-xr-x 1 root root 6912408 Jan 22 12:41 /usr/bin/mysq
```

User and group: User/Group: mysql / mysql, here its mysql instead of root. This is called Process Isolation. By running under its own "mysql" user instead of the "root".

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ps -ef | grep mysql
mysql        269    1    1 13:15 ?        00:00:07 /usr/sbin/mysqld
hpmuska+     572  373    0 13:26 pts/0    00:00:00 grep --color=auto mysql
```

Service management: MySQL runs as a Daemon (a background service). You control it using the systemctl utility.

To check status:

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ sudo systemctl statu
s mysql
[sudo] password for hpmuskansukhija:
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.ser>
     Active: active (running) since Wed 2026-03-25 14:>
    Process: 170 ExecStartPre=/usr/share/mysql/mysql-s>
   Main PID: 261 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 4534)
     Memory: 425.0M (peak: 438.3M)
        CPU: 2.263s
     CGroup: /system.slice/mysql.service
             └─261 /usr/sbin/mysqld

Mar 25 14:00:54 DESKTOP-3TC5UPL systemd[1]: Starting m>
Mar 25 14:00:57 DESKTOP-3TC5UPL systemd[1]: Started my>
```

to stop service:

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ sudo systemctl stop mysql
```

to start service:

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ sudo systemctl start mysql
```

to refresh:

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ sudo systemctl restart mysql
```

Update model: MySQL is open-source, you don't wait for a single company to mail you a DVD. Security patches are pushed through a global "Trust Network." When a bug is found, the open-source community fixes the code and pushes it to the Ubuntu Repositories.

To check for update:

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1551 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1846 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [249 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1166 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1665 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2701 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2837 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [627 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:25 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [656 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:30 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7364 B]
Get:31 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:33 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 15.4 MB in 59s (259 kB/s)
Reading package lists... Done
```

To apply patches:

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ sudo apt upgrade mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mysql-server is already the newest version (8.0.45-0ubuntu0.24.04.1).
Calculating upgrade... Done
The following packages will be upgraded:
  binutils binutils-common binutils-x86-64-linugnu cloud-init coreuti
  python3-software-properties software-properts-common systemd-hwe-hw
13 upgraded, 0 newly installed, 0 to remove and not upgraded.
Need to get 6442 kB of archives.
```

Part C – The FOSS Ecosystem

Dependencies- MySQL is a sophisticated piece of software that relies on several foundational open-source components which includes glibc, for core system functionality ,zlib and lz4 to handle data efficiency. It was primarily written in C and C++, requiring modern compilers like gcc or clang.

What MySQL has inspired- The most famous fork of MySQL is MariaDB which is created by MySQL's founder Michael Monty Widenius. Also, Persona Server, a fork focused on high performance maintains close compatibility with MySQL releases.

MySQL in LAMP stack- Mysql is the "M" in the LAMP stack, which stands for **L**inux, **A**pache, **M**ySQL, and **P**HP/Python/Perl. It serves as the **Data Persistence Layer**. While Linux is the foundation (OS), Apache is the gatekeeper (Web Server), and PHP is the logic (Scripting), MySQL is the "memory" where all user data, posts, and configurations are stored.for most of the popular content Management Systems (CMS) such as WordPress Joomla, this is the default database.

Community and governance- The governance of MySQL is owned and managed by Oracle Corporation. Its unique because it is a 'corporate-led' open-source project. Final decisions are made by oracle employees.

Developers and users primiraly use official mySQL forums to interact and the community meets at global conferences like persona live as well as my SQL user groups(MUGs).

Part D – Open Source Vs. Proprietary

Dimension	Open Source-MySQL	Proprietary-Oracle DB
Cost	Free to download	High cost(licencing fee per user)
Security	Transparent	Vendor controlled
Support and reliability	Community forums, slack	Paid professional vendor support

Freedom to modify	yes,you can modify code for own need	no, illegal to modify code
Community vs corporate Control	Community/oracle	Orace corp
Overall verdict	Suitable for users who are willing to create something new. They don't need to start from scratch.	Suitable for the company itself. Provides security to that proprietary.

In my view, MySQL works really well for web-based startups and applications that mostly involve reading data, like WordPress websites. It's fast, efficient, and most importantly, it doesn't have any entry cost, which makes it perfect for beginners and small teams.

However, when it comes to large financial systems that deal with complex queries, multiple table joins, and need strict ACID compliance, I would prefer Oracle Database. It's more powerful and better suited for handling such critical and high-demand environments.

If I had the chance to contribute, I would choose MySQL. The reason is simple—it powers a huge part of today's web. Even small improvements in its performance can help millions of developers and businesses who depend on it but can't afford expensive proprietary databases.

4. shell scripts

Script 1-System Identity report

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ cat system_identity.sh
#!/bin/bash
# Script 1: System Identity Report
# Author: Muskan | OSS Project

# --- Variables ---
STUDENT_NAME="Muskan"
SOFTWARE_CHOICE="MySQL"

# --- System Info ---
KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)
DISTRO=$(grep PRETTY_NAME /etc/os-release | cut -d= -f2 | tr -d '"')

# --- Display ---
echo "======================================"
echo " Open Source Audit - $STUDENT_NAME"
echo "======================================"
echo "Software : $SOFTWARE_CHOICE"
echo "Kernel   : $KERNEL"
echo "User     : $USER_NAME"
echo "Uptime   : $UPTIME"
echo "Date     : $DATE"
echo "Distro   : $DISTRO"
```

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ./system_identity.s
========================================
 Open Source Audit - Muskan
========================================
Software : MySQL
Kernel   : 6.6.87.2-microsoft-standard-WSL2
User     : hpmuskansukhija
Uptime   : up 12 minutes
Date     : Sat Mar 28 19:25:41 UTC 2026
Distro   : Ubuntu 24.04.4 LTS
License  : GPL (GNU General Public License)
========================================
```

Explanation- This script serves as a localized version "welcome screen" for the Linux environment. It uses variables to store student and software data and command substitution to get something on real-time system metrics like the kernel version via uname -r and uptime via uptime -p . The script also demonstrates string manipulation by parsing the /etc/os-release file to know the specific Linux distribution. This exercise highlights the open-source philosophy of transparency, as all system information is accessible by the command line.

Script 2-FOSS package inspector

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ cat package_inspector.sh
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="mysql"

# Check if package is installed (Debian-based)
if dpkg -l | grep -q "^ii  $PACKAGE"; then
    echo "$PACKAGE is installed."
    dpkg -s $PACKAGE | grep -E 'Version|License|Description'
else
    echo "$PACKAGE is NOT installed."
fi

# Case statement for description
case $PACKAGE in
    mysql)
        echo "MySQL: open source database powering millions of applicati
ons."
        ;;
    apache2)
        echo "Apache: web server that powers the open internet."
        ;;
    vlc)
        echo "VLC: open-source media player for all formats."
        ;;
    firefox)
        echo "Firefox: browser promoting an open web."
        ;;
    *)
        echo "Unknown package."
```

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ./package_inspector.sh
mysql is installed.
dpkg-query: package 'mysql' is not intalled and no information is ava
able
Use dpkg --info (= dpkg-deb --info) to examine archive files.
MySQL: open source database powering millions of applications.
```

Explanation- This script shows the package management and conditional logic. It uses an if-then-else structure to check for the presence of a software package example: MySQL also using system tools. A case statement is implemented to provide a "philosophy note" or description based on the package name, reasoning how shell scripts can categorize and interpret data. This aligns with Unit 2s focus on how Open-Source software is distributed and audited within a Linux ecosystem

Script 3-Disk and Permission auditor

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ cat disk_auditor.sh
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist on this system"
    fi
done

# Check MySQL config directory
MYSQL_DIR="/etc/mysql"

echo "----------------------"
if [ -d "$MYSQL_DIR" ]; then
    PERMS=$(ls -ld $MYSQL_DIR | awk '{print $1, $3, $4}')
    echo "MySQL Config Directory: $MYSQL_DIR => Permissions: $PERMS"
else
    echo "MySQL config directory not found."
fi
```

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ./disk_auditor.sh
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.4M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 104M
/home => Permissions: drwxr-xr-x root root | Size: 64K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 201M
/tmp => Permissions: drwxrwxrwt root root | Size: 20K
----------------------
MySQL Config Directory: /etc/mysql => Permissions: drwxr-xr-x root r
```

Explanation- The Disk and Permission Auditor focuses on file system hierarchy and security. It uses a for loop to iterate through an array of critical system directories. Within the loop, the script uses awk and cut to extract specific fields permissions and size from the output. By specifically auditing the MYSQL configuration directory, the script proves how Open-Source tools manage security through user and group ownership.

Script 4-Log File Analyzer

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ cat log_analyzer.sh
#!/bin/bash
# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

# Check if file exists
if [ ! -f "$LOGFILE" ]; then
    echo "Error: File $LOGFILE not found."
    exit 1
fi

# Check if file is empty
if [ ! -s "$LOGFILE" ]; then
    echo "File is empty. Please provide a valid log file."
    exit 1
fi

# Read file line by line
while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ./log_analyzer.sh /var/log/syslog
Keyword 'error' found 25 times in /var/log/syslog
Last 5 matching lines:
2026-03-27T16:00:59.796608+00:00 DESKTOP-3TC5UPL kernel: WSL (2 - init-
ystemd(Ubuntu)) ERROR: InitTerminateInstanceInternal:2708: systemctl po
eroff did not terminate the instance in 10000 ms, calling reboot(RB_POW
R_OFF)
2026-03-28T19:12:12.934568+00:00 DESKTOP-3TC5UPL systemd[1]: apport-aut
report.path - Process error reports when automatic reporting is enablec
(file watch) was skipped because of an unmet condition check (Conditior
athExists=/var/lib/apport/autoreport).
2026-03-28T19:12:12.934572+00:00 DESKTOP-3TC5UPL systemd[1]: apport-aut
report.timer - Process error reports when automatic reporting is enable
 (timer based) was skipped because of an unmet condition check (Conditi
nPathExists=/var/lib/apport/autoreport).
2026-03-28T19:12:12.940610+00:00 DESKTOP-3TC5UPL kernel: RAS: Correctab
e Errors collector initialized.
2026-03-28T19:12:16.788409+00:00 DESKTOP-3TC5UPL kernel: WSL (217) ERRO
: CheckConnection: getaddrinfo() failed: -5
```

Explanation- This script automates the task of log monitoring, a core skill is used in Linux administration. It uses while-read loop to process a log file line-by-line, combined with a counter variable to track occurrences of a specific keyword like- error. It handles command-line arguments for flexibility and includes an if-then check to ensure the file exists and is not empty before processing. The use of tail and grep at the end provides a quick snapshot of the most recent issues found.

Script 5-The Open Source Manifesto generator

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ cat manifesto_generator.sh
#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer the following questions:"
echo ""

read -p "1. Name one open-source tool you use: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date '+%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# Create manifesto
echo "On $DATE, I believe that open-source software like $TOOL represe
s $FREEDOM." > $OUTPUT
echo "I aim to contribute by building $BUILD and sharing it freely wit
the community." >> $OUTPUT

echo ""
echo "Manifesto saved to $OUTPUT"
echo "----------------------------------"
cat $OUTPUT
```

```
hpmuskansukhija@DESKTOP-3TC5UPL:~$ ./manifesto_generator.sh
Answer the following questions:

1. Name one open-source tool you use: MySQL
2. In one word, what does 'freedom' mean to you? unbounded
3. Name one thing you would build and share freely: deployment tool

Manifesto saved to manifesto_hpmuskansukhija.txt
----------------------------------
On 28 March 2026, I believe that open-source software like MySQL repre
nts unbounded.
I aim to contribute by building deployment tool and sharing it freely
th the community.
```

Explanation- This script emphasizes user interaction and file Input/Output. It uses the read command to capture user input interactively, which is then stored in variables. Through string concatenation, these inputs are woven into a personalized philosophy statement and written to a new .txt file using the output redirection operator > and append operator >>. This script demonstrates how automation can be used to generate documentation, showing the open-source values of sharing and community contribution.

5. A Closing Thought

The audit of MySQL highlights an interesting paradox in the open-source world: a project can be both managed by a corporation and still widely supported by the community. While some critics question its dual-licensing model as a commercial complication, the widespread use of MySQL shows that its core strengths—reliability and ease of use—continue to stand out.

Through this project, I've come to understand that open source is not just about being free of cost; it's about the longevity and ownership of code. Even if the company behind it no longer exists, the GPL license ensures that the code remains available to everyone.

Working on automation scripts during this audit has also strengthened my perspective as a future developer. It made me realize that contributing to the FOSS (Free and Open Source Software) ecosystem is not only the right thing to do ethically, but also an important step for professional growth.

6.References

On open source philosophy

- **The Open Source Initiative (OSI).**A foundational text defining the criteria for open-source software. https://opensource.org/osd
- **Free Software Foundation (FSF).** The Four Freedoms. Explains the ethical and philosophical difference between "Open Source" and "Free Software." https://www.gnu.org/philosophy/free-sw.html
- **Raymond, Eric S.** (1999).A seminal essay on open-source development models that influenced the evolution of projects like MySQL. http://www.catb.org/~esr/writings/cathedral-bazaar/

On Linux and shell scripting

- **The Linux Documentation Project (TLDP).**resource used for developing the logic for Scripts 1 through 5 in this audit. https://tldp.org/LDP/abs/html/
- **Free Software Foundation.** GNU Bash Manual. The official reference for the Bourne Again Shell (Bash) used in the automation scripts. https://www.gnu.org/software/bash/manual/

For finding license texts

- **MySQL Server License Header.** GitHub Repository - Oracle Corp. The specific file headers proving the use of the GPL-2.0 license in the MySQL source code. https://github.com/mysql/mysql-server/blob/8.0/README
- **Choose a License (by GitHub).** breakdown of the permissions, conditions, and limitations of the license used by MySQL. https://choosealicense.com/licenses/gpl-2.0/
- **The Software Package Data Exchange (SPDX).** License List: GPL-2.0-only. Provides the standardized identifier used in the Part B: Script 2 package inspection. https://spdx.org/licenses/GPL-2.0-only.html